

Valve Cyclers

Bench Tool User Guide

Ken Overton

KJO Liquid Rocket Motor Control System

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A standalone bench tool that repeatedly cycles a single servo-actuated valve between two PWM endpoints, for mechanical endurance/wear testing and for exercising a valve during bench assembly work.

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1 Purpose

Valve Cyclor drives one servo-actuated valve back and forth between two operator-supplied PWM counts (“Upper” and “Lower”), with a configurable pause between moves, either a fixed number of times or indefinitely. It is used for mechanical endurance testing, seating a newly-assembled valve, or simply exercising a valve on the bench without needing the full production firmware running.

2 Hardware Required

- Adafruit Feather M0 (ATSAMD21G18A)
- Adafruit 8-channel PWM/Servo FeatherWing (PCA9685), I²C
- One servo-actuated valve, servo wired to **PCA9685 channel 7**
- USB cable for Serial (115200 baud)

Note

The servo channel (7) is hardcoded in the sketch. To cycle a valve wired to a different channel, edit `SERVO_CHANNEL` in `main.cpp` and reflash.

3 How to Use

Open a Serial monitor at 115200 baud after powering the board. On every power-up (and again after a run completes) the tool asks four questions, each with a **y/n** confirmation before moving to the next:

1. **Upper** – the maximum PWM count (one travel endpoint).
2. **Lower** – the minimum PWM count (the other travel endpoint).
3. **Delay D** – seconds to hold at each endpoint before moving to the other.
4. **Number of cycles** – how many Lower→Upper→Lower round trips to run, or **−1** for indefinite cycling.

If you answer **n** to any confirmation, the tool re-asks that same question. Once all four are confirmed, cycling starts immediately: the servo moves to Lower, waits **D** seconds, moves to Upper, waits **D** seconds, and repeats. A finite run prints "**N cycles completed.**" and returns to the setup questions for another run; an indefinite run (**−1**) never stops on its own – power-cycle the board to end it.

4 Notes

- The Serial console has no timeout – the tool waits indefinitely at every prompt.
- Backspace is supported while typing a value.
- Upper and Lower are used exactly as entered – the tool does not check that Upper is actually greater than Lower, or that either value is within the servo’s safe travel range. Use the companion **Servo Tuner** bench tool first to find safe endpoints for a given valve.

5 Version History

Version	Date	Notes
0.1	September 2026	Initial user guide.